

MTN

PROJECT 01 · QUANTITATIVE & PREDICTIVE RISK MODELLING

MTN QuantRisk Intelligence Platform

AI-powered news scraping, risk detection & prediction for MTN Ghana and MTN Group

PROJECT START

18 May 2026

MILESTONE 1

15 Jun 2025

FINAL DEADLINE

29 Jun 2026

Executive Summary



15+

News Sources Monitored

500+

Articles per Day

6

Risk Categories

30d

Forecast Horizon

Project Overview

What it does

Scrapes 15+ global and Ghana-specific news sources every 15 minutes, runs fine-tuned AI models to detect MTN-relevant risks, and forecasts 7-day and 30-day financial impact in GHS & USD.

Web App First

React dashboard with real-time risk radar, heat maps, predictive charts, and configurable alert thresholds — live by 15 June 2025.

Then Mobile

React Native app (iOS & Android) plus installable PWA with push notifications, biometric login, and offline caching — complete by 29 June 2025.

AI at the Core

BERT risk classifier, spaCy NER pipeline, LSTM forecaster, Monte Carlo simulation, and LLM-generated board-ready briefings.

Project Timeline

6 WEEKS · MAY 18 – JUN 29, 2026

Milestone 1
Jun 15

Final
Jun 29

Wk 1
May 18

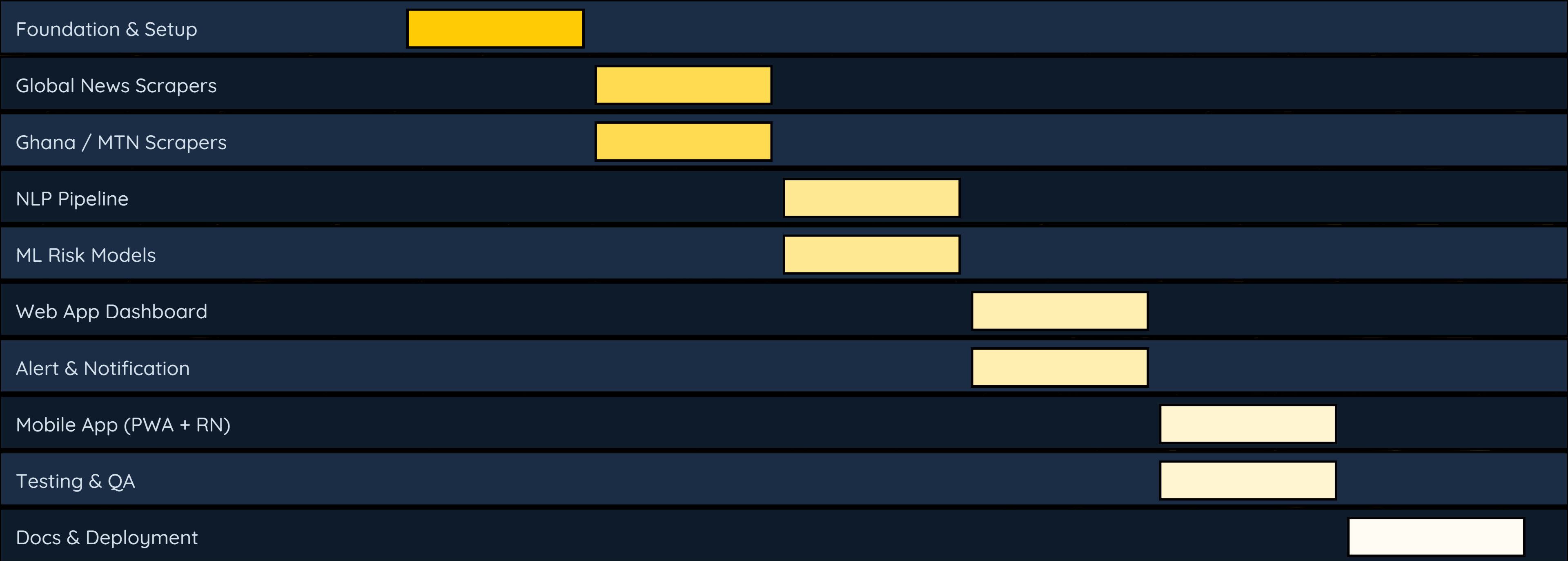
Wk 2
May 25

Wk 3
Jun 1

Wk 4
Jun 8

Wk 5
Jun 15

Wk 6
Jun 22



Infrastructure

Data & Scraping

ML & NLP

Web App

Mobile

Deploy & QA

01

Foundation & Architecture

MAY 18-24 · WEEK 1

Repo & DevOps

GitHub + Docker + CI/CD pipeline via GitHub Actions. Containerised from day one.

News Source Audit

Inventory 15+ sources, verify robots.txt compliance, define scraping strategy per domain.

API Blueprint

FastAPI endpoints mapped out. JWT authentication scheme documented.

Risk Taxonomy

Six risk categories: Regulatory, Competitive, FX, Operational, Political, Reputational.

Database Design

PostgreSQL schema for articles & risk scores. Redis cache layer for real-time feed serving.

Architecture Doc

Full system diagram, data-flow map, and technology decisions signed off by team.

✓ DELIVERABLE

Architecture doc + skeleton codebase committed to GitHub

02

Web Scrapers & Data Pipeline

MAY 25-31 · WEEK 2

Global Finance Scrapers

Reuters, Bloomberg, Financial Times, AP, BBC. Scrapy spiders with Playwright for JS-rendered pages.

MTN-Specific Sources

MTN Investor Relations, NCA Ghana releases, GSMA reports, MTN Ghana earnings calls.

Task Scheduler

Celery beats every 15 min. Deduplication engine prevents re-processing identical articles.

Ghana Local Scrapers

JoyFM, CitiFM, Graphic Online, GhanaWeb, Business Ghana direct MTN relevance.

Social Media Pipeline

Twitter/X API monitoring #MTN, #MTNGhana, #Ghana. Reddit & LinkedIn sentiment stream.

Pipeline Capacity

Target: 500+ articles/day across all sources with <2 min latency from publish to ingestion.

✓ DELIVERABLE

Live pipeline ingesting 500+ articles/day across 15+ sources

03

NLP & ML Risk Models

JUN 1-7 · WEEK 3

spaCy NER Pipeline

Entity extraction: MTN mentions, GHS/USD amounts, regulations, competitor names, people.

Sentiment Model

Article-level positive / neutral / negative scoring calibrated for finance & telecom domain.

LSTM Forecaster

Time-series model trained on 12 months of historical MTN news for 7-day & 30-day projections.

Fine-tuned BERT Classifier

MTN-relevance scoring + 6-category risk tagging. Target F1 > 0.85 on held-out test set.

Impact Scoring Model

Estimates financial impact range in GHS and USD based on article risk profile and entity weight.

Anomaly Detection

Flags unusual clustering of negative news for immediate escalation to Watch/Warning/Critical.

✓ DELIVERABLE

Trained models with F1 > 0.85 + full evaluation report

04

Web App Dashboard

JUN 8-14 · WEEK 4 · MILESTONE 1: JUN 15

Live Risk News Feed

React 18 frontend displaying AI-scored MTN news in real time via WebSocket push.

Predictive Charts

7-day and 30-day risk trend forecasts with Monte Carlo min/mid/max impact bands.

Alert & Notification System

Three tiers: Watch / Warning / Critical. Email via SendGrid, in-app push via WebSocket.

MTN Risk Radar

Interactive heat map across Ghana regions and global MTN markets. Powered by D3.js & Recharts.

Smart Drill-Down Filters

Filter by risk category, geography, severity level, date range instant in-browser query.

AI Narrative Reports

LLM generates plain-English board-ready one-page risk briefs from model outputs automatically.

✓ DELIVERABLE

Web App v1.0 deployed and publicly accessible Milestone 1 achieved

05

Mobile App

JUN 15-21 · WEEK 5

PWA Wrapper

Installable on any device from the browser. Works offline with cached latest risk summaries.

Push Notifications (FCM)

Firebase Cloud Messaging delivers Watch/Warning/Critical alerts even when app is closed.

Biometric Login

Face ID / Fingerprint authentication via Expo LocalAuthentication for secure quick access.

React Native App

Cross-platform iOS and Android build targeting real devices via Expo managed workflow.

Mobile Risk Dashboard

Full risk feed, heat map thumbnails, and predictive charts optimised for small screens.

Data Sync

Shared REST + WebSocket API keeps mobile fully in sync with web backend in real time.

✓ DELIVERABLE

Mobile app on TestFlight (iOS) and Play Store beta (Android)

06

Testing, Docs & Deployment

JUN 22-29 · WEEK 6 · FINAL DEADLINE: JUN 29

End-to-End Test Suite

Cypress for frontend flows + Pytest for backend. Full regression on all critical paths.

Model Back-Testing

Validate LSTM forecaster against 12 months of historical MTN news data.

Production Deployment

AWS EC2 + RDS + ElastiCache behind Nginx reverse proxy. Let's Encrypt SSL certificate.

Load Testing

Simulate 1,000 concurrent users on AWS. Identify and fix bottlenecks before go-live.

Technical Documentation

API reference, deployment runbook, architecture diagrams, and operator user guide.

Final Presentation

Slide deck, live demo, and executive summary report ready for stakeholder review.

✓ DELIVERABLE

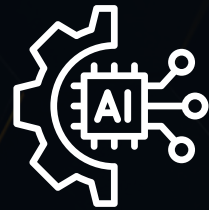
Full platform live + final presentation ready — Project Complete

News Sources & Data Inputs

15+ MONITORED SOURCES · EVERY 15 MINUTES

Ghana & Africa

- JoyFM / myjoyonline.com
- CitiFM / citinewsroom.com
- Graphic Online
- GhanaWeb
- Business Ghana
- TechCabal / Quartz Africa



Global Finance & Telecom

- Reuters
- Bloomberg
- Financial Times
- Associated Press / BBC
- Wall Street Journal
- GSMA Intelligence



Social & Sentiment

- Twitter/X API (#MTNGhana)
- Reddit r/ghana r/africa
- LinkedIn MTN Group
- YouTube MTN Ghana
- Google Trends (Telecom)



Regulatory & MTN-Specific

- NCA Ghana (press releases)
- ICASA South Africa
- MTN Group Annual Reports
- MTN Investor Relations
- ITU News
- MTN Ghana Earnings Calls



Platform Features

EIGHT CORE CAPABILITIES

01

Multi-Source Web Scraper

15+ sources crawled every 15 minutes. Global finance, Ghana local, MTN investor & social media in one unified pipeline.

02

AI Risk Classifier

Fine-tuned BERT reads every article, scores MTN-relevance, tags six risk categories with confidence score & financial impact.

03

Real-Time Risk Dashboard

Interactive web app with live risk radar, heat maps by geography & risk type, Monte Carlo projections, 30-day forecasts.

04

Smart Alert System

Configurable threshold alerts: Watch / Warning / Critical. Delivered via in-app push, email, and mobile notification.

05

Predictive Risk Forecasting

LSTM time-series model forecasts 7 and 30 days ahead. Monte Carlo gives min/mid/max financial impact in GHS & USD.

06

Mobile App + Offline Mode

React Native (iOS & Android) + installable PWA. Push alerts, biometric login, and offline caching of latest risk summaries.

07

Ghana + Global Intelligence

Dual-layer: Ghana-specific risks (NCA, GHS/USD FX, political calendar) + global MTN Group risks (macro, investor sentiment).

08

AI Narrative Reports

LLM generates plain-English board-ready one-page briefs automatically from model outputs and news clusters.



MTN QuantRisk Intelligence Platform

Quantitative & Predictive Risk Modelling · Project 01

***From breaking news to board-ready briefings
in under 15 minutes.***

6 Weeks

End-to-End
Delivery

15 Min

News-to-Alert
Latency

2 Apps

Web + Mobile
Platform